

GEFEI (ESTHER) LIU

gefei_liu@brown.edu | <https://estherliu02.github.io/>

EDUCATION

• Brown University

Master of Science, Computer Science (Research Track)

Sep 2025 - Apr 2027

Providence, Rhode Island

◦ GPA: 4.0/4.0

◦ Advisors: Prof. Serena Booth, Prof. Michael Littman

• University of Waterloo

Bachelor of Computer Science, Major in Data Science, Minor in Statistics

Sep 2020 - Apr 2025

Waterloo, Ontario

◦ GPA: 90.24% (Dean's Honours)

PUBLICATIONS AND PREPRINTS

- [1] Gefei Liu, Sonya Rashkovan, Sophia Lloyd George, Isaac Sheidlower, and Serena Booth, **Ai safety via debate is compromised by cognitive biases**, under review, 2026 [\[paper\]](#).
- [2] Mingxi Jia, Patton Yin, Haojie Huang, Ivy Xiao He, Xuanru Liu, **Gefei Liu**, Dian Wang, and Stefanie Tellex, **Learning Fine-Grained Manipulation from Human Videos via Dual-Actor Policy**, under review, 2026 [\[paper\]](#).
- [3] Weimin Wu, Alexander C Furnas, Eddie Yang, **Gefei Liu**, Akhil Pandey Akella, Xuefeng Song, Dashun Wang, and Han Liu, **Sci2Pol: Evaluating and Fine-tuning LLMs on Scientific-to-Policy Brief Generation**, in *ICLR*, 2026 [\[paper\]](#).
- [4] **Gefei Liu**, Tian-Ying Wu, Chun Kit Kwok, and Juewen Liu, **Computational Analysis Uncovering Contrasts in G-Quadruplex Propensity and Aptamer Enrichment in SELEX-Derived Libraries**, *ChemBioChem*, vol. 27, no. 1, e202500614, 2026 [\[paper\]](#).
- [5] Weimin Wu, Zhihan Zhou, Robert Riley, Mohammad Abdulqader, Xuefeng Song, Satria Kautsar, Rob Egan, Steven Hofmeyr, **Gefei Liu**, Shira Goldhaber-Gordon, Mutian Yu, Harrison Ho, Yaping Liu, Andrei Stecca Steindorff, Fengchen Liu, Feng Chen, Rachael Morgan-Kiss, Lizhen Shi, Han Liu, and Zhong Wang, **Uncovering the Genomic Manifold via Scalable Learning from the Global Microbiome**, *bioRxiv*, 2025 [\[paper\]](#).

RESEARCH AND TEACHING EXPERIENCES

• BrownAI: A Campus Information Chatbot for Brown University Undergraduates

Jul 2026 - Present

supervised by [Professor Michael Littman](#) at Brown University

◦ Conducting formative and user studies, using JEV and thematic coding to inform chatbot design.

• A Corpus-Grounded Interface for Student Writing Revision

Feb 2026 - Present

supervised by [Professor Serena Booth](#) at Brown University and [Professor Elena Glassman](#) at Harvard University

◦ Designing corpus-grounded interfaces that use prior human-written examples as evidence for feedback to improve student-teacher-system interaction and the quality and efficiency of writing feedback.

• A Dashboard for an Internet Censorship Observatory

May 2024 - Aug 2024

supervised by [Professor Diogo Barradas](#) at University of Waterloo

[\[Code: 📄\]](#)

◦ Optimized a web-based visualization dashboard for exploring [ICLab](#) Internet censorship measurements by country and measurement type, improving usability, deployment, and database performance.

• Enhancing a Main-Memory DBMS

Sep 2022 - Dec 2022, Jan 2024 - Apr 2024

supervised by [Professor Grant Weddell](#) at University of Waterloo

◦ Developed ARM schema specifications and SQLP query examples for a main-memory DBMS, supporting abstract entity identifiers, navigational foreign-key references, and compact structured-data integration.

• Tutor for MATH237, Calculus 3 for Honours Mathematics

Jan 2022 - Apr 2022

University of Waterloo Mathematics Tutoring Centre

OTHER PROJECTS

- **DyTac: Dynamic Touch Sensing Without Tactile Sensors** Spring 2026
CSCI2951K: Collaborative Robotics, instructed by [Professor Stefanie Tellex](#) [Report: [🔗](#)]
 - Developed a temporal proprioceptive sensing framework that localizes whole-body touch on a moving SPOT robot without tactile sensors, using real-world and simulated locomotion data.
- **LLM Honesty via Debate: Are Honest Models More Persuasive?** Fall 2025
CSCI1952A: Human-AI Interaction, instructed by [Professor Serena Booth](#) [Report: [🔗](#)]
 - Investigated whether honesty improves persuasiveness in AI debate by fine-tuning LLMs with varying honesty levels, generating structured debate transcripts, and comparing LLM and human judgments.
- **Contrastive Subspace Steering for LLM Safety** Fall 2025
CSCI2952X: Research Topics in Self-Supervised Learning, instructed by [Professor Randall Balestriero](#) [Report: [🔗](#)]
 - Developed a lightweight activation-steering method that learns unsafe subspaces from safe-unsafe response pairs to reduce LLM harmfulness while preserving helpfulness without model finetuning.
- **Time Series Forecasting** Spring 2024
STAT443: Forecasting, instructed by [Jeremy Vanderdoes](#) [Report: [🔗](#)]
 - Applied statistical time-series models across hydrological, financial, production, and pollution data, evaluating forecasts with cross-validation, prediction intervals, and model diagnostics.
- **Noisy Label Learning for Medical Image Classification** Winter 2024
CS486: Introduction to Artificial Intelligence, instructed by [Professor Jesse Hoey](#) [Report: [🔗](#)]
 - Reviewed five state-of-the-art noisy-label learning methods for medical image classification regarding their robustness, scalability, computational trade-offs, and limitations under varying noise levels.

INTERNSHIPS

- **Data Analyst @ Capital One [🌐](#)** Sep 2024 - Dec 2024
 - Productionized and validated enterprise analytics datasets using Snowflake, Git, and AWS, while automating data workflows and supporting business teams with governed, production-ready insights.
- **Data Management Analyst @ Environment and Climate Change Canada [🌐](#) [Code: [🔗](#)]** Jan 2024 - Apr 2024
 - Built FAIRER data assessment tools, restored a 500+ term research data management glossary through web recovery, and extended machine-actionable data management plan standards using HTML, CSS, JavaScript, Python, and Go.
- **Data Analytics Consultant @ BDO Canada [🌐](#)** Jan 2023 - Apr 2023
 - Led QA testing for [SmartMine](#), optimized Power BI dashboards, and supported Azure-based data integration to improve data quality, dashboard performance, and operational insights.
- **Data Migration Analyst @ CN Railway [🌐](#)** May 2022 - Aug 2022
 - Supported SAP HANA database migration by cleansing 1M+ records and optimizing SQL queries to reduce duplication, improve retrieval speed, and shorten project timelines.

AWARDS

- **Bradley Family Upper-Year Scholarship for Women in Mathematics** 2024
awarded annually to a woman in Mathematics with research in data analytics, coding, computer science, or artificial intelligence
- **University of Waterloo President's Research Award** 2024
awarded for A Dashboard for an Internet Censorship Observatory supervised by Diogo Barradas
- **University of Waterloo President's Research Award** 2023
awarded for Enhancing a Main-Memory DBMS supervised by Grant Weddell
- **University of Waterloo President's Scholarship of Distinction** 2021
awarded for an admission average of 95% or above

ADDITIONAL INFORMATION

Languages: English (Bilingual Proficiency), Mandarin (Primary Proficiency), Spanish (IB Spanish Standard Level 6), Korean (Diploma in Korean Language I by University of Waterloo)

GRE: Verbal: 164, Quantitative: 170, Writing: 4.5

References: [Prof. Serena Booth](#), [Prof. Michael Littman](#), [Prof. Elena Glassman](#), [Prof. Han Liu](#)